# Assignment No- 04

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Course - CSE320

Section - 20

## Ans to the Q No-1

To send 'Hello' we have to use hexadecimal values for checksum

Using ascii table -

Sender site:

| | | |
|---|---|---|
| 48 | 65 | (He) |
| 6C | 6C | (ll) |
| 00 | 6F | (o) |
| 00 | 00 | |
| B5 | 40 | |

$\downarrow$ 1's compliment

4A BF (checksum)

Our checksum is 4ABF. Now let's ~~code~~ check it

it has error or not.

P.T.O.

Receiver side :

48 65 (He)
6C 6C (ll)
00 6F (o)
4A BF (recived checksum)

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FF FF

$\big\downarrow$ 1's compliment

$\boxed{00\ 00}$

As all the values are zero it means our check sum is correct,

To ensure my enemy know if the message they recieved is correct we have to send ~~both~~ the message and the checksum. So in hexadecimal the message will be : 4865 6C6C 006F 4ABF.

(Ans)

# Ans to the Q No-2

Direct sequence spread spectrum or DSSS is a modulation technique that spreads the signal over a wider bandwidth than the orisginal information signal.

**Advantages of DSSS:**

i) DSSS can provide improved security through the use of spreading codes because here 1 bit convert to n bit which is though to crack.

ii) DSSS can support higher data rates compare to other spread sprectrum technique.

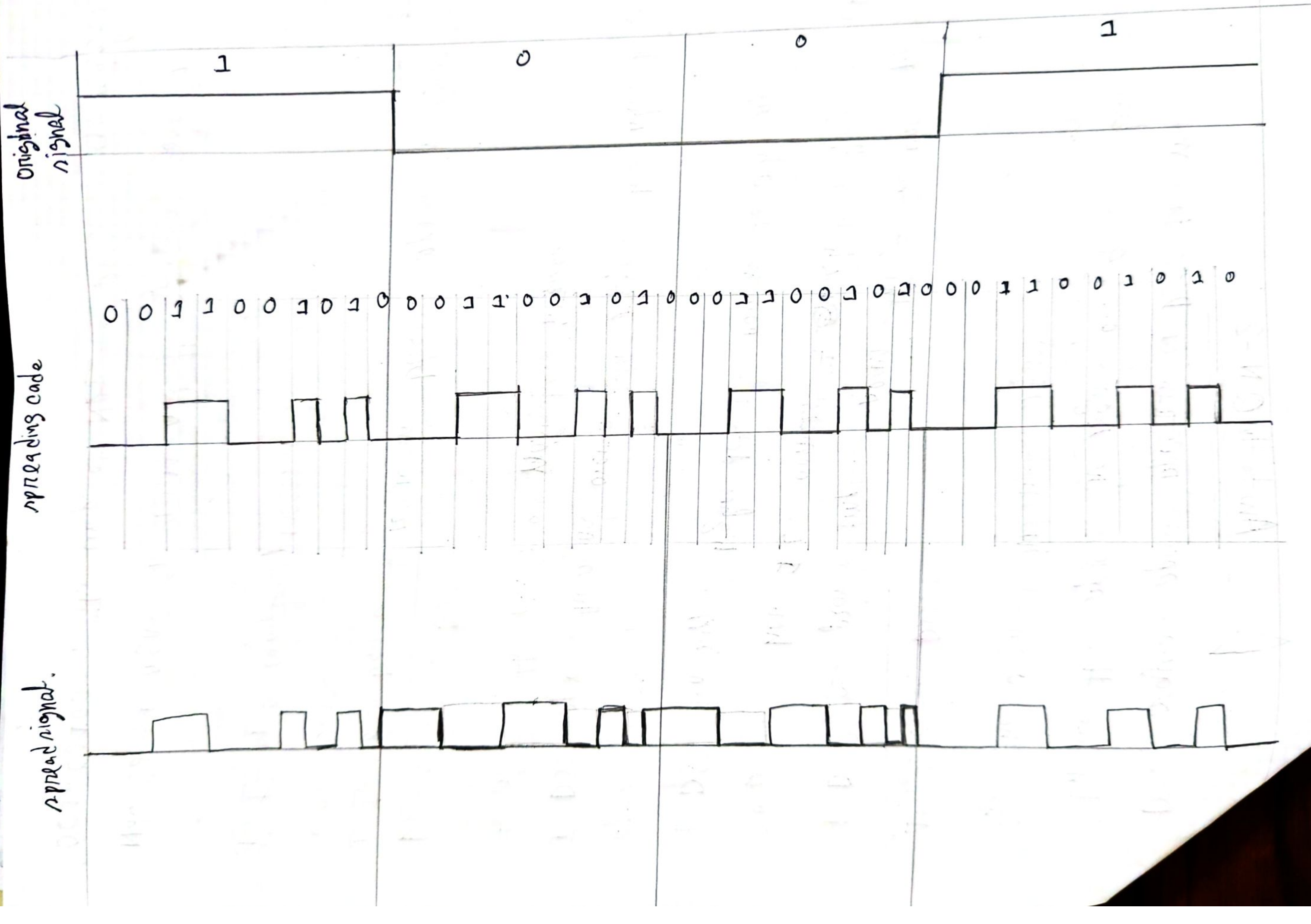
iii) DSSS spreads the signal over a wide frequency band which helps to mitigate the effects of narrowband interference.

**Disadvantages of DSSS:**

i) Increases bandwidth requirement because it spreads the signal over wider bandwidth.

ii) It is a complex process.

Now if the signal is 1001 and the spreading code is 0011001010. The unipolar NRZ and DSSS is given below.


1
0
0
1
Original signal
0 0 1 1 0 0 1 0 1 0 0 0 1 1 0 0 1 0 1 0 0 0 1 1 0 0 1 0 1 0 0 0 1 1 0 0 1 0 1 0
spreading code
spread signal.

# Ans to the Q No- 3

Carrier Sense Multiple Access with collision detection (CSMA/CD) is a MAC protocol which is commonly used for guided media such as ethernet. ~~and also for radio waves particularly like wireless networks like wifi~~ Ethernet is wired which is LAN so it is also CSMA/CD. Radio waves is commonly used in ALOHA. particularly in wireless networks like wifi.

We know,

$$T_B = \text{Random number} \times T_p \quad (\text{Propagation time})$$

This random value is between 0 to $2^k-1$. Here k is the number of attempts. So if the number of attempt increases the $T_B$ will also increase.

Here,

propagation time, $T_p = \frac{40}{13000}$ s

$= 3.076 \times 10^{-3}$ s.

$= 3.08$ ms.

Random number for 5th attempt $=$ 0 to $(2^5 - 1)$

$=$ 0 to 31.

So,

Back off time $T_B = ($random number $\times 3.08)$ ms.

So,

$T_B = (0 \times 3.08)$ ms $= 0$ ms.

$T_B = (1 \times 3.08)$ ms $= 3.08$ ms

$T_B = (2 \times 3.08)$ ms $= 6.16$ ms.

⋮

$T_B = (31 \times 3.08)$ ms $= 95.48$ ms.

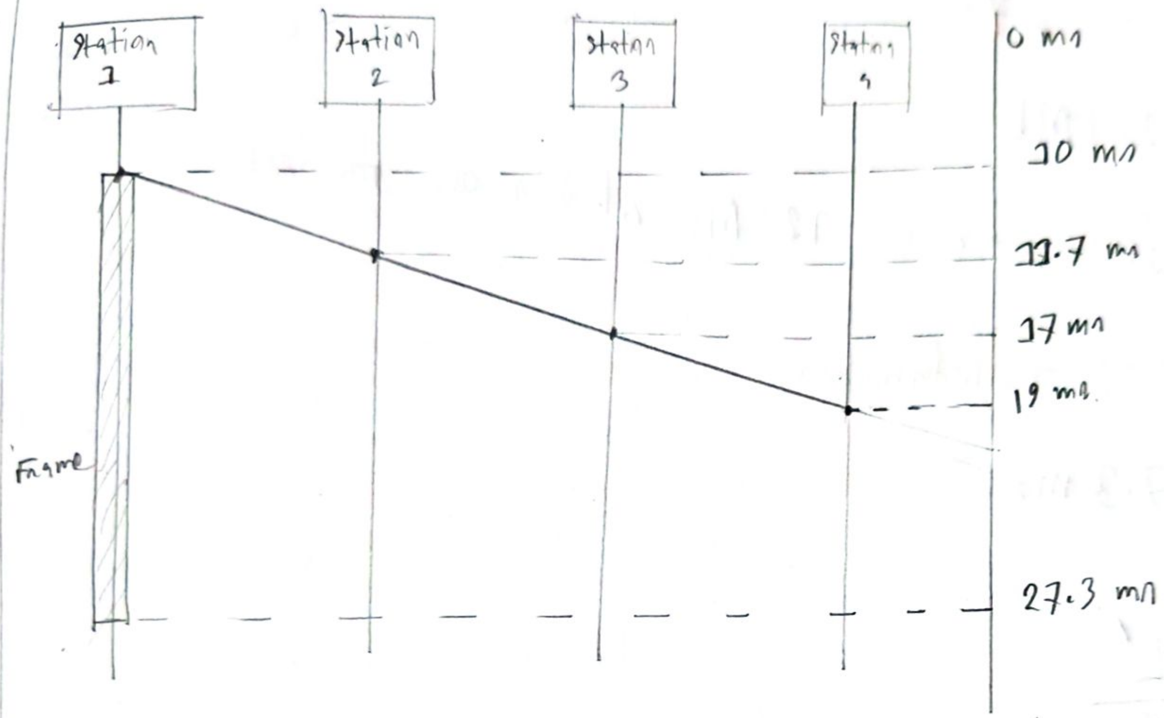
So the back off time will be ~~between 0 ms to 95.48 ms.~~

~~between~~ 0 ms, 3.08 ms, 6.16 ms, .... 95.48 ms.

(Ans)

# Ans to the Q No-4

space and time model of the channel if it is using CSMA/CD is given—




Maximum propagation time, $T_P = (19 - 10)$ ms

$= 9$ ms

Transmission time, $T_{fr} = (27.3 - 10)$ ms

$= 17.3$ ms.

## Standard frame Size:

$T_{fr} = 2 \times T_P$

$= 2 \times 9$

$= 18$ ms.

Bandwidth given 7 Gbps.

Now,

$$T_{fr} = \frac{L}{B}$$

$$\Rightarrow L = T_{fr} \times B$$

$$\Rightarrow L = 18\times10^{-3} \times 7\times10^{9}$$

$$\Rightarrow L = 126 \text{ Mb}$$

minimum frame size is 126 Mb which is our standard.

Now according to information,

$$T_{fr} = 17.3 \text{ ms.}$$

Now,

$$T_{fr} = \frac{L'}{B}$$

$$\Rightarrow L' = 17.3\times10^{-3} \times 7\times10^{9}$$

$$\Rightarrow L' = 121.1 \text{ Mb.}$$

Here, $126 > 121.1$ so, $L > L'$ which means our minimum standart is greater that our actual frame size.

So the answer will be no. It doesnot meet the requirements as it is smaller that the standard frame size.